

Quiz-2: Deep Learning

14th March, 2026

Name: _____ Roll No: _____

Subject Code: ☐ AI2100 ☐ AI5100 ☐ CS5480

Duration: 90 minutes

Total : 50 Marks.

Please read the following comments before you work on the problems:

- Please neatly write your name, roll number, and subject code in the space provided on the question paper.
- Write your solutions clearly within the designated space for each question. Answers written in illegible or cluttered handwriting may not be evaluated.
- Ensure that you return the complete question paper to the invigilator once the exam ends.
- This is a closed-book quiz. Calculators, phones, or any programmable devices are strictly prohibited. Any form of cheating or plagiarism, whether detected during the exam or later during grading, will result in **F grade in the course**.

1 Multiple Choice Questions (Single Correct)

[1 marks × 13 = 13 marks]

1. Alice and Bob are developing CNN architectures to perform image classification. Their architectures are identical except for the set of operations performed post Global Average Pooling (GAP): Alice flattens the output and then applies a dense layer. Bob instead performs a 1×1 convolution on the GAP output, followed by flattening, but does not apply a dense layer after. Assume that an ideal CNN classifier will learn to classify the dataset being considered. Then, select the correct statement from the ones below:
 - A. Alice's network will learn to classify the dataset, while Bob's network will not.
 - B. Bob's network will learn to classify the dataset, while Alice's network will not.
 - C. Both Bob's and Alice's networks will learn to classify equally well.
 - D. Neither Alice's nor Bob's networks will learn to classify well.
2. In a Convolutional Neural Network, what is the primary reason for using the Rectified Linear Unit (ReLU) activation function instead of the Sigmoid function?
 - A. ReLU is more computationally expensive, leading to better accuracy.
 - B. ReLU helps mitigate the vanishing gradient problem during backpropagation.
 - C. ReLU constrains the output of each neuron to be between -1 and 1 .
 - D. ReLU is only used in the final fully connected layer of the network.
3. Which of the following describes a key difference in how convolution and pooling operations treat the channel dimension C ?
 - A. Both operations convolve across channel dimensions.

- B. Convolution preserves number of channels, while pooling always reduces them to a single representative map.
- C. Pooling is applied independently to each channel, whereas convolution combines information across all input channels.
- D. Convolution uses non overlapping stride in the channel dimension, while pooling uses an overlapping one.
4. What property does pooling provide that makes a CNN somewhat robust to small translations in the input image?
- A. Sparse Connectivity. B. Weak Invariance. C. Equivariance. D. Weight sharing.
5. What is the result of the 1D convolution operation: $(0, 0, 1, 1, 0, 0.1, 0.2, 1, 1, 1, 0) \otimes (1, 1)$
- A. $(0, 0, 1, 2, 1, 0.1, 0.3, 1.2, 2, 2, 1, 0)$
- B. $(0, 1, 1, 0, 0.1, 0.2, 1, 1, 1, 0)$
- C. $(1, 2, 2, 1.1, 0.3, 1.2, 2, 2, 1, 0)$
- D. None of the above.
6. In the standard VGG-style pooling layer with a 2×2 pool size and stride 2, how does the total number of floating-point operations (FLOPs) change if you double the output channels after pooling?
- A. The computational cost is halved.
- B. The computational cost increases by a factor of 4.
- C. The computational cost is doubled.
- D. The computational cost remains unchanged.
7. For an input of size $H \times W$ and a kernel of size $h \times w$ with no padding and stride 1, what is the correct formula for the output dimensions?
- A. $(H - h + 1) \times (W - w + 1)$.
- B. $(H - h) \times (W - w)$.
- C. $(H + h - 1) \times (W + w - 1)$.
- D. $H/h \times W/w$.
8. When utilizing Cross-Entropy loss $H(p, q) = -\sum_i p_i \log(q_i)$ for a single sample with a ground-truth class c , which statement accurately describes the loss behavior as the predicted probability q_c approaches 0?
- A. The gradient of the loss becomes zero, leading to a plateau in learning.
- B. The loss increases logarithmically toward infinity, creating a strong penalty for high-confidence incorrect predictions.
- C. The loss converges to 1 because the target distribution p is one-hot encoded.
- D. The loss remains bounded by the maximum value of the one-hot encoded vector.
9. What is the consequence of initializing all weights W in a Deep Neural Network to zero?
- A. All neurons in a layer will perform the same calculation and receive the same gradient, preventing them from learning different features.
- B. The model will converge to the global minimum in the first epoch due to the balanced starting point.
- C. It forces the use of a very high learning rate η to move away from the origin.
- D. The activation functions will always saturate, leading to exploding gradients.
10. The Xavier initialization strategy suggests setting $W = \text{np.random.randn}(d_i, d_{i-1}) / \sqrt{d_{i-1}}$. What is the primary theoretical goal of this normalization?
- A. To reduce the memory footprint of the weight matrices during the forward pass.
- B. To ensure that all weights are positive to avoid dead ReLU neurons.
- C. To eliminate the need for any data preprocessing or input normalization.
- D. To keep the variance of the activations and gradients approximately constant across the layers of the network.
11. When applying Parameter Norm Penalties (L_1 or L_2 regularization), why are the bias terms b typically left unregularized?
- A. Biases are handled automatically by the learning rate schedule and do not need further constraints.
- B. The gradient of the bias term is always zero, so regularization would have no mathematical effect.

- C. Biases have fewer parameters than weights and regularizing them significantly increases bias without substantially reducing variance.
- D. Regularizing biases would lead to a division by zero in the Softmax layer.
12. The left subfigure in Figure 1 shows a black-and-white image, while the right subfigure shows the corresponding image with edges highlighted and non-edge pixels suppressed. In other words, pixels containing edges in the left subfigure are assigned significantly higher values in the right subfigure than pixels without edges. Which of the following kernels is likely to produce the right subfigure when convolved with the left subfigure?

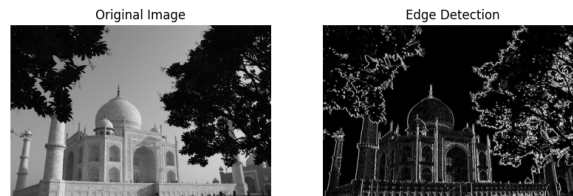


Figure 1: Edge Detection using Convolution Kernel

- A. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ C. $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$ D. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
13. Which of the following statements about optimization algorithms typically used for Deep Learning is TRUE?
- A. In the SGD+Momentum algorithm, a low friction coefficient (e.g., around 0.1) will typically cause the iterates to escape very fast from a saddle point or local minima.
- B. In Ada Grad, parameters with smaller gradient norm receive updates more frequently, as the features corresponding to those parameters occur more frequently (and therefore, in general, the objective ensures that the norm is anyway small for those parameters).
- C. Assume that a neural network is learning some ERM objective, under the same initialization and learning rate schedules. Further, assume that training runs for one whole epoch. Then if mini-batch SGD processes each training sample exactly once during an epoch (in any order), then the expected parameter vector after one epoch of mini-batch SGD is identical to the parameter vector obtained after one step of Batch Gradient Descent computed on the entire dataset (where batch size is the whole training set size).
- D. Assume that a neural network is learning some ERM objective, under the same initialization and a constant learning rate schedule. Further, assume that training runs for one whole epoch. Then, even if SGD processes each training sample exactly once during an epoch (in any order) with learning rate η with no noise added to the gradients, and Batch Gradient Descent computed on the entire dataset (where batch size is the whole training set size) uses learning rate $\frac{\eta}{N}$ where N is the training dataset size, the expected parameter vector obtained from running both optimizers would differ.

2 Multiple Choice Questions (Multi Correct)

[2 marks×7=14 marks]

1. Consider training a deep feedforward network using gradient descent and select all the TRUE statements.
- A. Initializing all weights to zero prevents learning because gradients are zero everywhere.
- B. Initializing all weights to the same nonzero constant prevents different neurons in a layer from learning distinct features.
- C. Very small random initialization can cause activations to shrink toward zero in deeper layers, leading to vanishing gradients.
- D. Xavier initialization attempts to keep both forward activations and backward gradients at similar variance across layers.
- E. The introduction of techniques like Batch Normalization generally reduces a deep network's sensitivity to the exact choice of weight initialization.

2. Assume a deep neural network with linear layers and zero-mean weights. Which of the following statements are correct?
 - A. If weight variance is too small, activations tend to collapse toward zero as depth increases.
 - B. If weight variance is too large, activations can grow exponentially with depth.
 - C. Keeping activation variance constant across layers also stabilizes gradient variance in backpropagation.
 - D. Xavier initialization ensures exact preservation of variance for any nonlinearity.
 - E. Xavier initialization assumes independence and zero mean of inputs and weights.
3. Consider the receptive field growth in a CNN when stacking L layers of 3×3 convolutions with a stride of 1. Which of the following statements are correct?
 - A. Receptive Field grows linearly with L .
 - B. Receptive Field grows exponentially with L .
 - C. Receptive field after L layers is $(2L + 1) \times (2L + 1)$.
 - D. Same receptive field can be achieved using one large kernel but fewer non-linearities.
 - E. Stacked small kernels introduce more parameters than one large kernel.
4. Which of the following statements correctly describes how the choice of architecture (ANN vs. CNN) affects the number of trainable parameters when processing an input image?
 - A. In an ANN, increasing the input image resolution significantly increases the number of trainable parameters in the first layer.
 - B. In a CNN, the number of trainable parameters in a convolutional layer is independent of the input image height and width.
 - C. Applying an ANN to an image results in fewer parameters than a CNN because it does not use filters.
 - D. In a CNN, the number of trainable parameters is primarily determined by the filter size, the number of filters, and the number of input channels.
5. Tick the correct statements regarding Convolutional Neural Networks (CNNs):
 - A. Skip connections can only be used in CNNs; they cannot help with the vanishing gradient problem in very deep Dense networks.
 - B. In a CNN, using a stride greater than one in the convolutional layers can help reduce the computational load and the size of the feature maps.
 - C. Dilated convolutions in CNNs are used to increase the receptive field without increasing the number of parameters.
 - D. When using MaxPooling, gradients flow even through the features that are discarded (pooled out).
6. Two students, *Marga* and *Neya*, are building a system to detect illegal mining activity from 64×64 RGB satellite patches. Mining sites often leave thin bright trails and irregular excavation textures that may appear anywhere in the image and may vary with lighting, terrain, vegetation, and sensor conditions. Training data comes primarily from Region A, but the system will be deployed in Region B, where conditions differ somewhat. *Marga* and *Neya* therefore suspect domain shift may occur.

Marga: "Since MLPs are universal approximators, we can flatten the $64 \times 64 \times 3$ image and train a sufficiently large MLP."

Neya: "A CNN should work better because it can exploit spatial structure and detect local patterns such as trails."

Because they do not have large compute resources, they cannot train very deep models and must design careful experiments to test these hypotheses. Which of the following experiments would be appropriate for *Neya* to design to test whether CNNs are more suitable for this task?

- A. Train a baseline MLP on flattened images and a small CNN (3–4 convolutional layers) on the same dataset, keeping the number of parameters roughly comparable, and compare validation accuracy.
- B. Evaluate both models on images where the mining artefacts appear at spatial locations different from those seen during training.
- C. Train using data augmentation (e.g., random translations and brightness changes) to simulate variations in lighting and environment, and compare how each model benefits.
- D. Randomly shuffle the pixels in every image before training both models, and if the CNN still performs

better conclude that CNNs are superior.

E. Train a very large MLP until it achieves near-zero training loss, and if its training accuracy exceeds the CNN's conclude that the MLP is the better architecture.

7. Suppose that a function $f : \mathbb{R}^4 \rightarrow \mathbb{R}$ is *can be* completely learned by a parametrized network, $\tilde{f}_W : \mathbb{R}^4 \rightarrow \mathbb{R}$ that does the following: it first convolves the input vector $x \in \mathbb{R}^4$ with a parametrized weight vector $W \in \mathbb{R}^5$, and then sums up the maximum of the first 4 and last 4 elements of the convolved sequence. Further, suppose this is to be trained by computing the mean-squared error loss (L) between given training signals for the given training data points and updating W via vanilla gradient descent. Lastly, assume that W has been initialized as $[0, 0, 0, 0, 0]$. Then, which of the following are true regarding the optimization involved in the above scenario?

- A. It is a non-convex problem in the parameters W , even though convolution is a linear operation and mean-squared error is a convex function.
- B. If we removed the max-pooling operation, then the resulting optimization problem would have unique global minimum for W .
- C. If we were solving this problem by using vanilla gradient descent, then to choose the learning rate each time an update on W is made, we can solve the one-dimensional convex problem $\eta^* = \arg \min_{\eta} \tilde{f}_W \left(W_t + \eta \frac{\partial L}{\partial W_t} \right)$ and update W_t using η^* .
- D. At a particular iterate of vanilla gradient descent, finding the optimal learning rate $\eta^* = \arg \min_{\eta} \tilde{f}_W \left(W_t + \eta \frac{\partial L}{\partial W_t} \right)$ is itself a non-convex problem with many local minima and may require methods such as AdaGrad or Adam to solve it.

3 Fill in the blanks

[9 marks]

- The output of convolving the sequence $[8, 0, -1, 5, 3]$ with $[2, 16, -5, 4]$, in that order, is the sequence _____. [2 marks]
- Augmenting the SGD update equation with _____ can help it potentially jump out of local minima, reduce jitter over ravines, and make updates smoother. [1 mark]
- Consider a fully-connected layer $y = Wx$ where $W \in \mathbb{R}^{512 \times 256}$. Then the value of a for which initializing the network with Xavier-Glorot initialization is equivalent to initializing it with the uniform initialization, $w_{ij} \sim U(-a, a)$, is _____. [2 marks]
- In a residual block, a skip connection is added so that learning the _____ function becomes easier with deeper layers. [1 mark]
- Min-pooling is a _____ [linear/non-linear] operation and convolution is a _____ [linear/non-linear] operation, so the composed operation consisting of applying min-pooling over convolved inputs is a _____ [linear/non-linear] operation. [3 marks]

4 Subjective Questions

[13 marks]

- Consider the convolutional neural network shown in Figure 2. The input is a 6×6 grayscale image X . A single 3×3 convolution filter with bias and stride = 1 is applied to produce a feature map Z_1 , followed by a ReLU activation producing A_1 . A 2×2 max-pooling layer with stride = 2 is then applied to obtain P_1 . The resulting feature map is flattened into a vector F and passed to a fully connected neuron (FNN) with **no hidden layer** to produce Z_2 . A sigmoid activation is applied to obtain the final output $\hat{y}$. The network is trained for binary classification using the binary cross-entropy (BCE) loss.

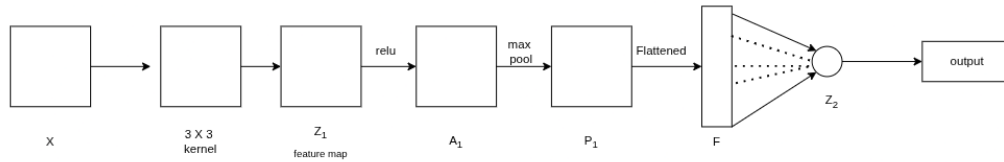


Figure 2: CNN architecture used in the question.

Answer the following:

- (a) Determine the **dimensions** of the following blocks: [2 marks]

$$Z_1 =$$

$$A_1 =$$

$$P_1 =$$

$$F =$$

$$Z_2 =$$

$$\hat{y} =$$

- (b) Compute the **total number of trainable parameters** in the network (including all weights and biases). [1 marks]

- (c) Using **backpropagation**, derive the gradients of the loss with respect to:

- the fully connected layer parameters (W_2, b_2) [2 marks]
- the convolution filter parameters (W_1, b_1)

[4 marks]

Clearly indicate the intermediate variables used during backpropagation.

2. Consider a toy multilayer perceptron $\mathcal{F}_\theta$ with:

[4 marks]

- Input dimension: 2;
- Two hidden layers, each with hidden dimension: 2;
- Activation: identity for each layer (linear network);
- Loss: Mean Squared Error (MSE);
- Optimizer: Vanilla Gradient Descent with learning rate $\eta = 0.5$.

The learning task consists of learning the function $f^* \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = x_1^2 + x_2^2$, over the input distribution $\mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)$ via empirical risk minimization on the MLP $\mathcal{F}_\theta$.

The training data consists of the two samples $x^{(1)} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $x^{(2)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, drawn from a standard Gaussian for which the corresponding ground-truth targets under f^* are $f^*(x^{(1)})$, $f^*(x^{(2)})$ respectively.

The network parameters are initialized as:

$$W_1 = \begin{bmatrix} 1 & 0 \\ 2 & 5 \end{bmatrix}, \quad W_2 = \begin{bmatrix} 1 & -1 \\ 3 & 1 \end{bmatrix}, \quad W_3 = \begin{bmatrix} 1 & -1 \end{bmatrix}.$$

Accordingly, at initialization, the network $\mathcal{F}_\theta$ is $W_3(W_2(W_1x))$.

Now let $W_j^{[t]}$ denote the value of the weight matrix just after the t^{th} update to W_j , for any $j \in \{1, 2, 3\}$. By convention, we define $t = 0$ as the initial values of the parameters before any application of vanilla gradient descent.

Next, we define the internal covariate shift $\Delta_j(t)$, at layer j , between timesteps t and $t + 1$ as $D_{\text{KL}} \left(W_j^{[t+1]}(W_{j-1}^{[t+1]} \dots (W_1^{[t+1]}x)) \parallel W_j^{[t]}(W_{j-1}^{[t]} \dots (W_1^{[t]}x)) \right)$ for inputs x sampled from the distribution specified in the learning task.

Lastly, define $\Delta_j^{\text{BN}}(t)$ to be the internal covariate shift $\Delta_j(t)$, at layer j , between timesteps t and $t + 1$ as for $\Delta_j(t)$, except that a Batch-Normalization layer is now inserted in between W_2 and W_1 in $\mathcal{F}_\theta$ with initialization of parameters γ, β as both 1.

Then compute the following numerical quantities with clear derivation¹:

1. $\Delta_2^{\text{BN}}(0) - \Delta_2(0)$;
2. $\Delta_3^{\text{BN}}(0) - \Delta_3(0)$.

¹You may use the following facts:

1. γ, β are scale and shift parameters respectively. Furthermore, you may take $\epsilon = 0$.
2. $D_{\text{KL}}(\mathcal{N}(\mu_{1d \times 1}, \Sigma_{1d \times d}) \parallel \mathcal{N}(\mu_{2d \times 1}, \Sigma_{2d \times d})) = \frac{1}{2} \left[\ln \left(\frac{\det(\Sigma_2)}{\det(\Sigma_1)} \right) + (\mu_2 - \mu_1)^T \Sigma_2^{-1} (\mu_2 - \mu_1) - d + \text{trace}(\Sigma_2^{-1} \Sigma_1) \right]$.